

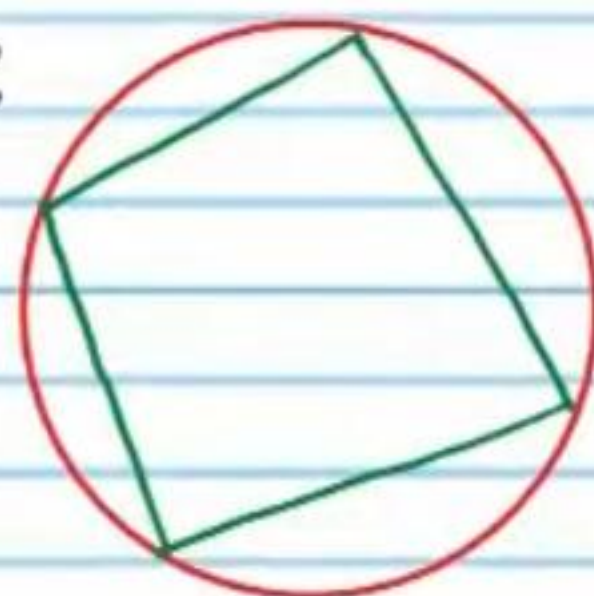
Area (Part III): Regular Polygons, Regular Pyramids, and Geometric Probability

Area of Regular Polygons

Inscribed Polygon: A polygon whose vertices are located on a circle.

Circumscribed Circle: The circle around an inscribed polygon.

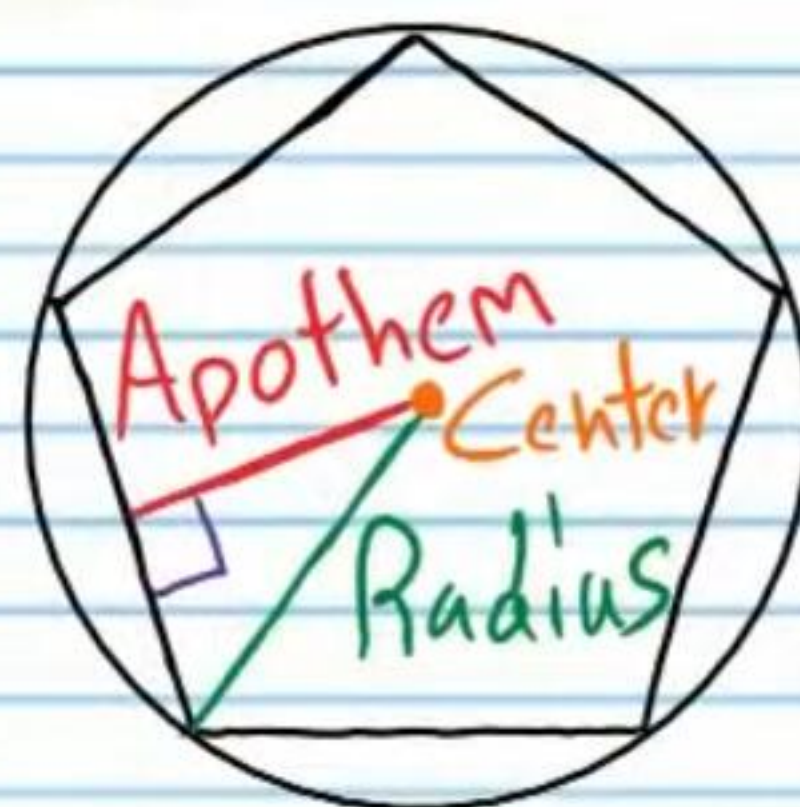
Example:



Center of a Regular Polygon: The center of the circle circumscribed around a regular polygon.

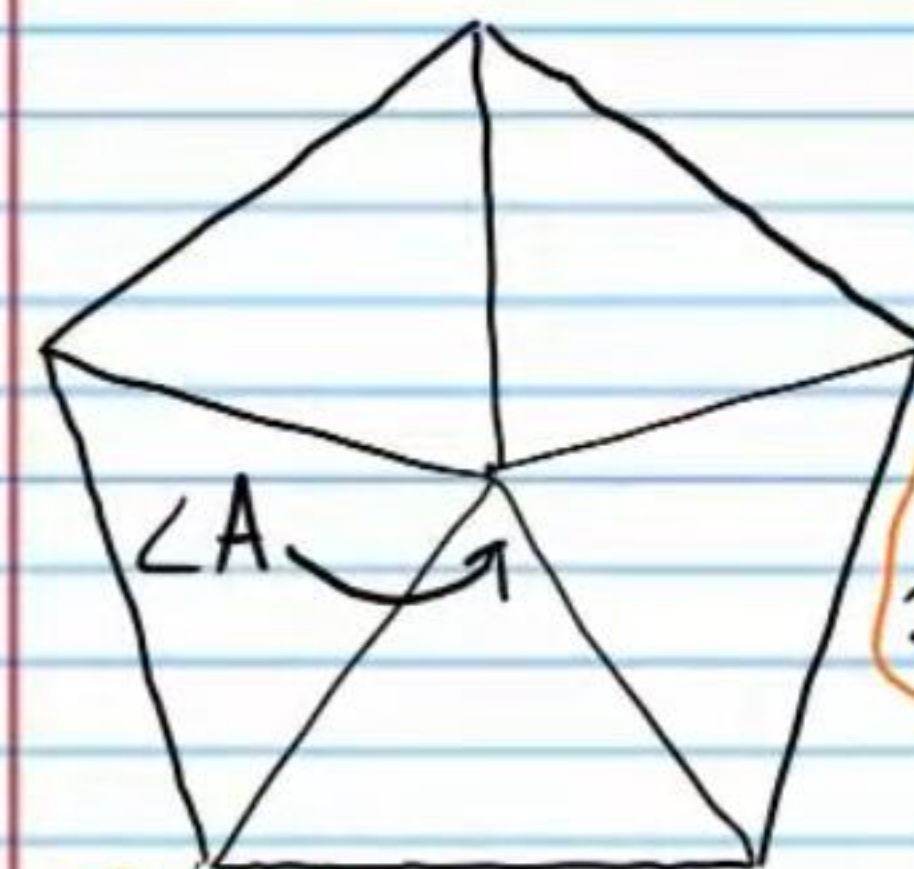
Radius of a Regular Polygon: The distance from the center to a vertex.

Apothem: A segment extending from the center of a regular polygon to a side and perpendicular to the side. The word also refers to the distance from the center to a side.



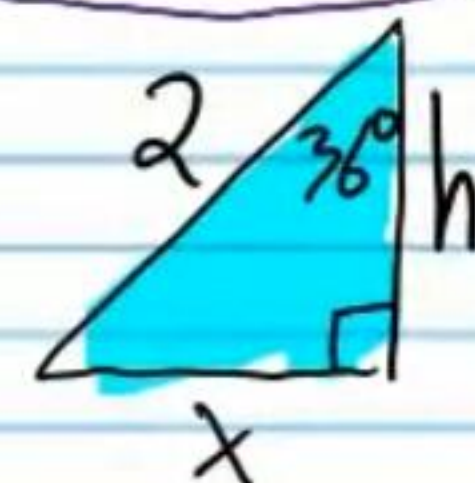
Find the area of each regular polygon below with the given characteristics. Round to the nearest hundredth.

① Radius: 2 m



$$\text{I) } m\angle A = \frac{360}{5} = 72^\circ$$

$$\text{II) Height} \\ \cos 36^\circ = \frac{h}{2} \\ 2 \cos 36^\circ = h$$



III) Base

$$\sin 36^\circ = \frac{x}{2}$$

$$2 \sin 36^\circ = x$$

$$b = 2x$$

$$b = 2(2 \sin 36^\circ)$$

$$b = 4 \sin 36^\circ$$

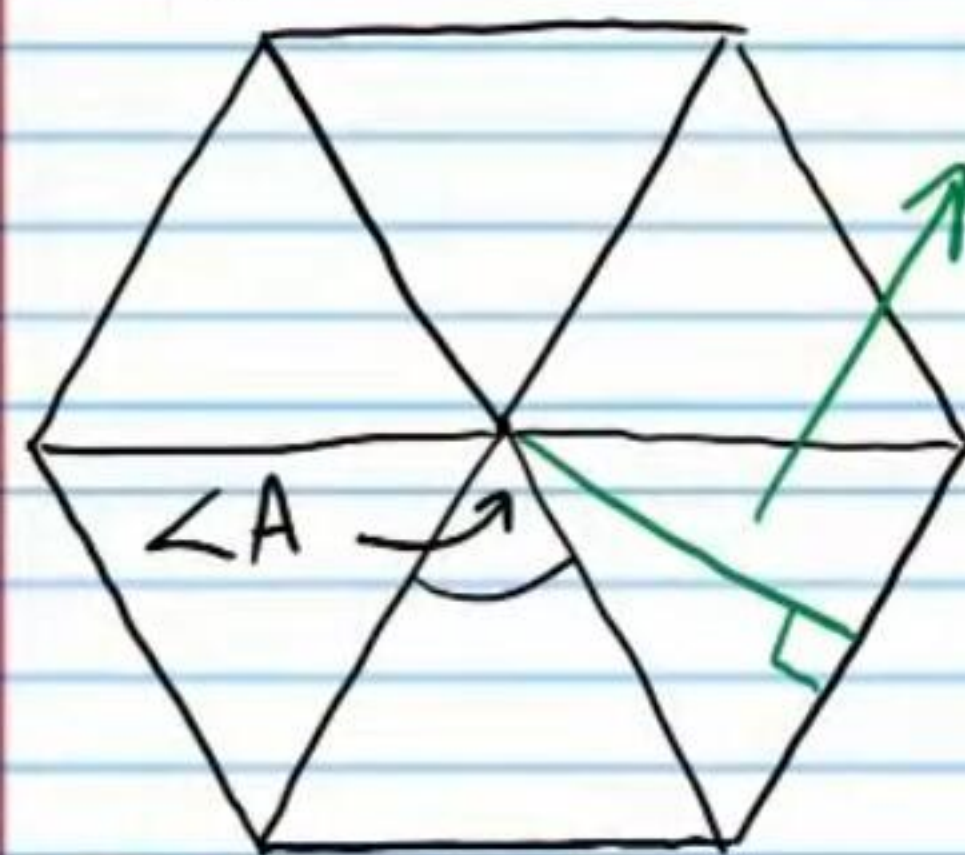
$$\text{IV) } A_{\text{tri}} = \frac{1}{2} (4 \sin 36^\circ) (2 \cos 36^\circ)$$

$$\text{V) } A_{\text{total}} = 5 A_{\text{tri}}$$

$$A_{\text{total}} = 5 \left[\frac{1}{2} (4 \sin 36^\circ) (2 \cos 36^\circ) \right]$$

$$A_{\text{total}} \approx 9.51 \text{ m}^2$$

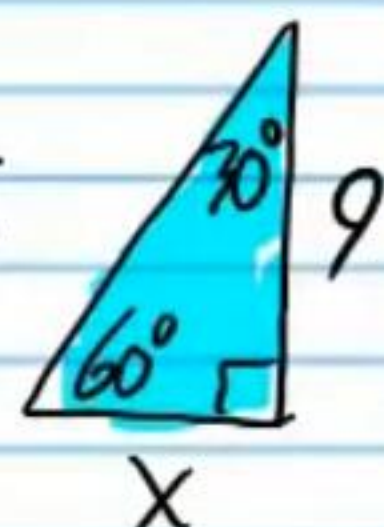
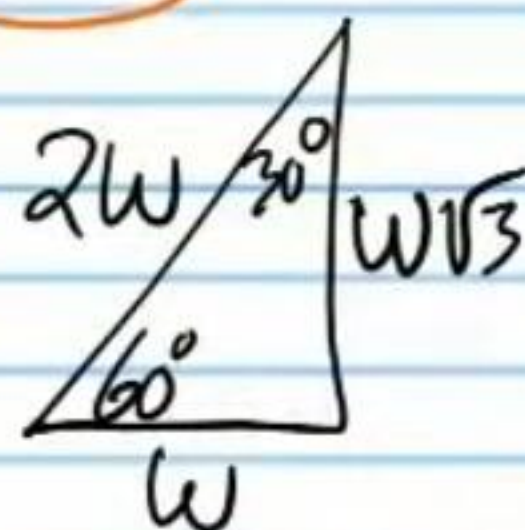
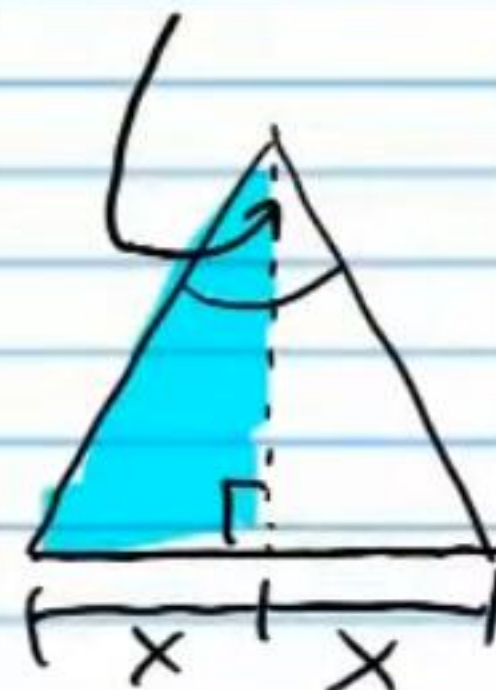
② Apothem: 9 mm



I) $m\angle A = 360/6 = 60^\circ$

II) 9 mm

Height
 $h = 9$



III) Base

$$w\sqrt{3} = 9$$

$$w = \frac{9}{\sqrt{3}}$$

$$x = w = \frac{9}{\sqrt{3}}$$

$$b = 2x$$

$$b = 2\left(\frac{9}{\sqrt{3}}\right)$$

$$b = \frac{18}{\sqrt{3}}$$

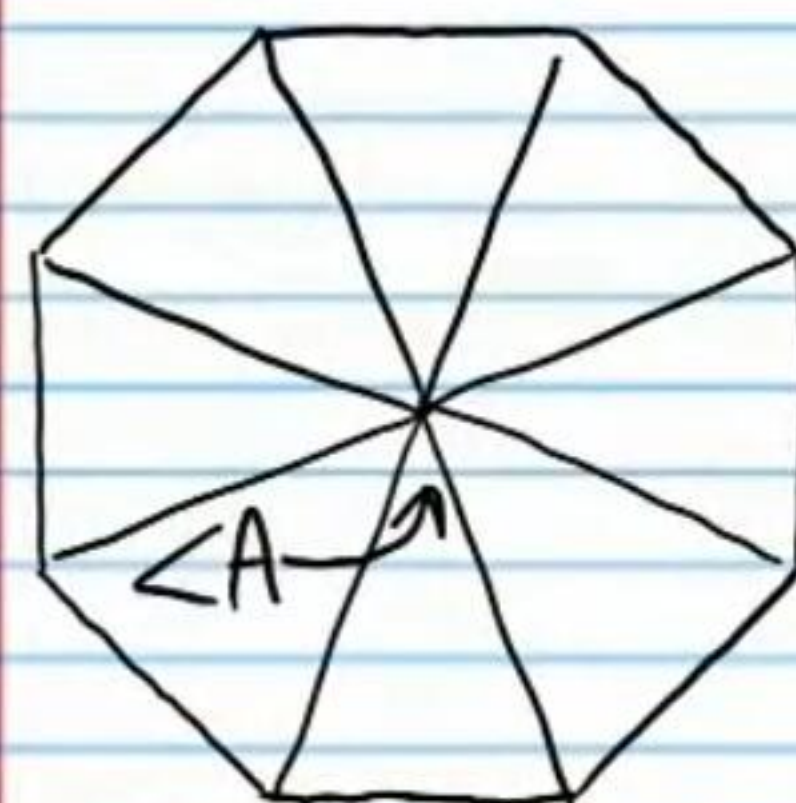
IV) $A_{\text{Tri}} = \frac{1}{2}\left(\frac{18}{\sqrt{3}}\right)(9)$

V) $A_{\text{Total}} = 6A_{\text{Tri}}$

$$A_{\text{Total}} = 6\left[\frac{1}{2}\left(\frac{18}{\sqrt{3}}\right)(9)\right]$$

$$A_{\text{Total}} \approx 280.59 \text{ mm}^2$$

③ Side Length: 7 ft



I) $m\angle A = 360/8 = 45^\circ$

II) Base
 $b = 7$



III) Height

$$\tan 22.5^\circ = \frac{3.5}{h}$$

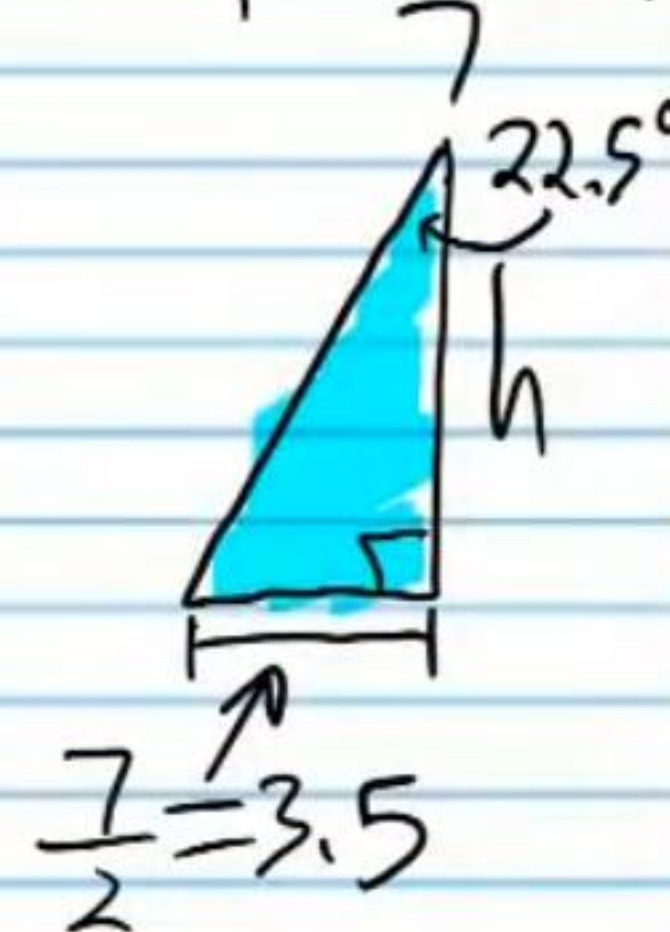
$$h = \frac{3.5}{\tan 22.5^\circ}$$

IV) $A_{\text{Tri}} = \frac{1}{2}(7)\left(\frac{3.5}{\tan 22.5^\circ}\right)$

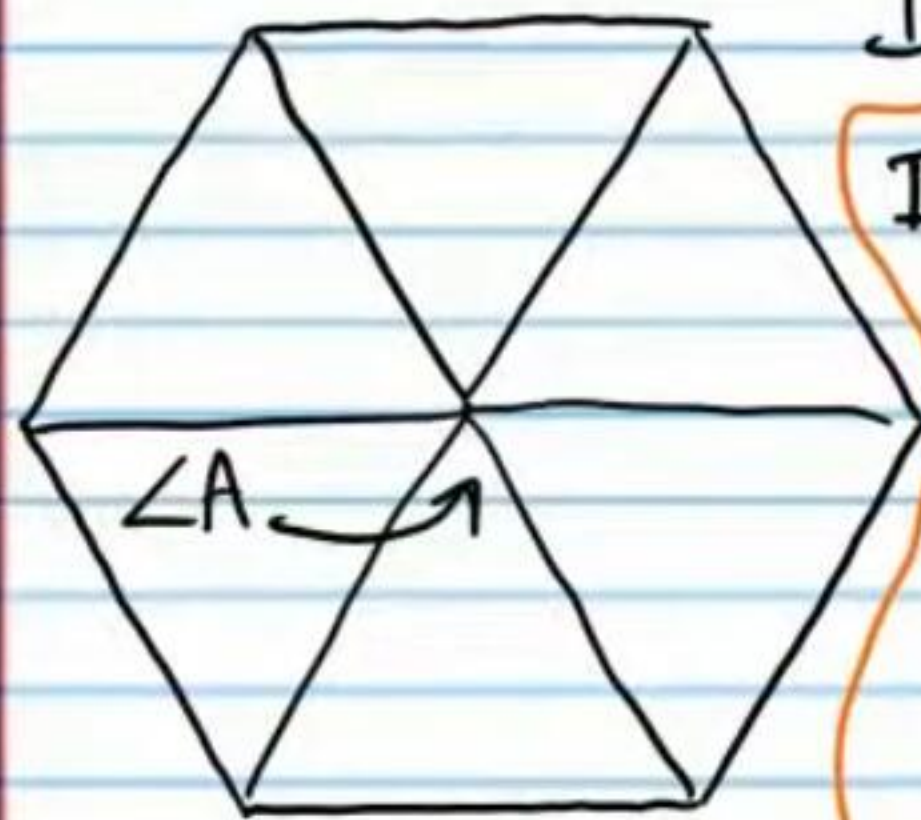
V) $A_{\text{Total}} = 8A_{\text{Tri}}$

$$A_{\text{Total}} = 8\left[\frac{1}{2}(7)\left(\frac{3.5}{\tan 22.5^\circ}\right)\right]$$

$$A_{\text{Total}} \approx 236.59 \text{ ft}^2$$



④ Radius 3 cm.



I) $m\angle A = 360/6 = 60^\circ$

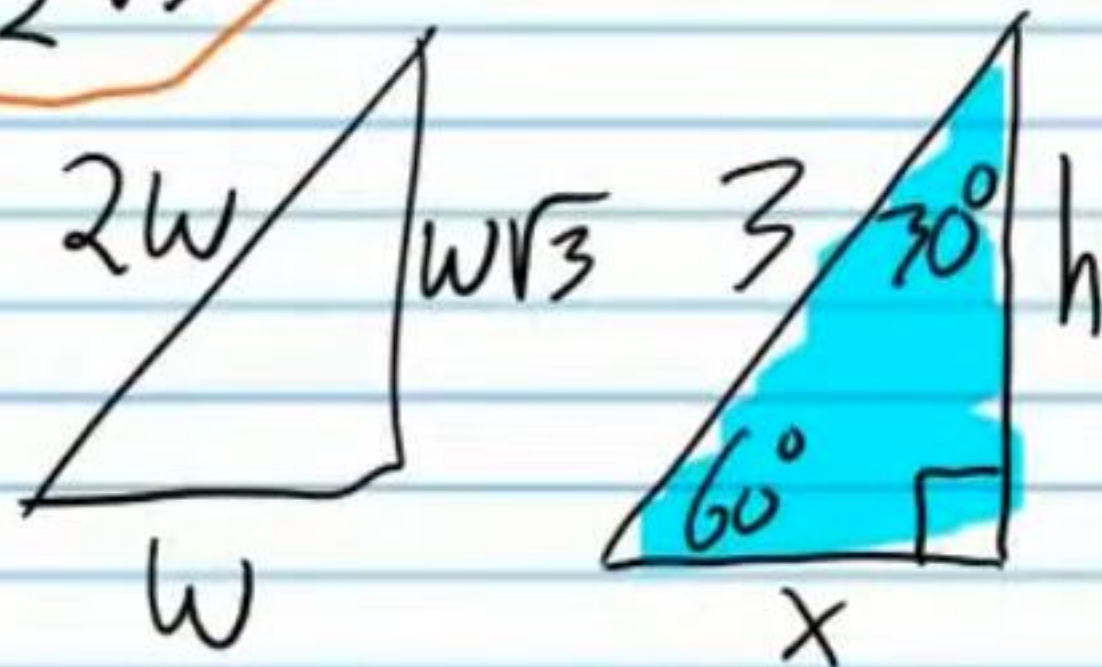
II) Height

$$2w = 3$$

$$w = \frac{3}{2}$$

$$h = w\sqrt{3}$$

$$h = \frac{3}{2}\sqrt{3}$$



III) Base

$$x = w = \frac{3}{2}$$

$$b = 2x$$

$$b = 2\left(\frac{3}{2}\right)$$

$$b = 3$$

$$IV) A_{Tri} = \frac{1}{2}(3)\left(\frac{3}{2}\sqrt{3}\right)$$

$$V) A_{Total} = 6\left[\frac{1}{2}(3)\left(\frac{3}{2}\sqrt{3}\right)\right]$$

$$A_{Total} \approx 23.38 \text{ cm}^2$$

⑤ Apothem: 4 in



I) $m\angle A = \frac{360^\circ}{7}$

II) Height

$$h = 4$$

III) Base

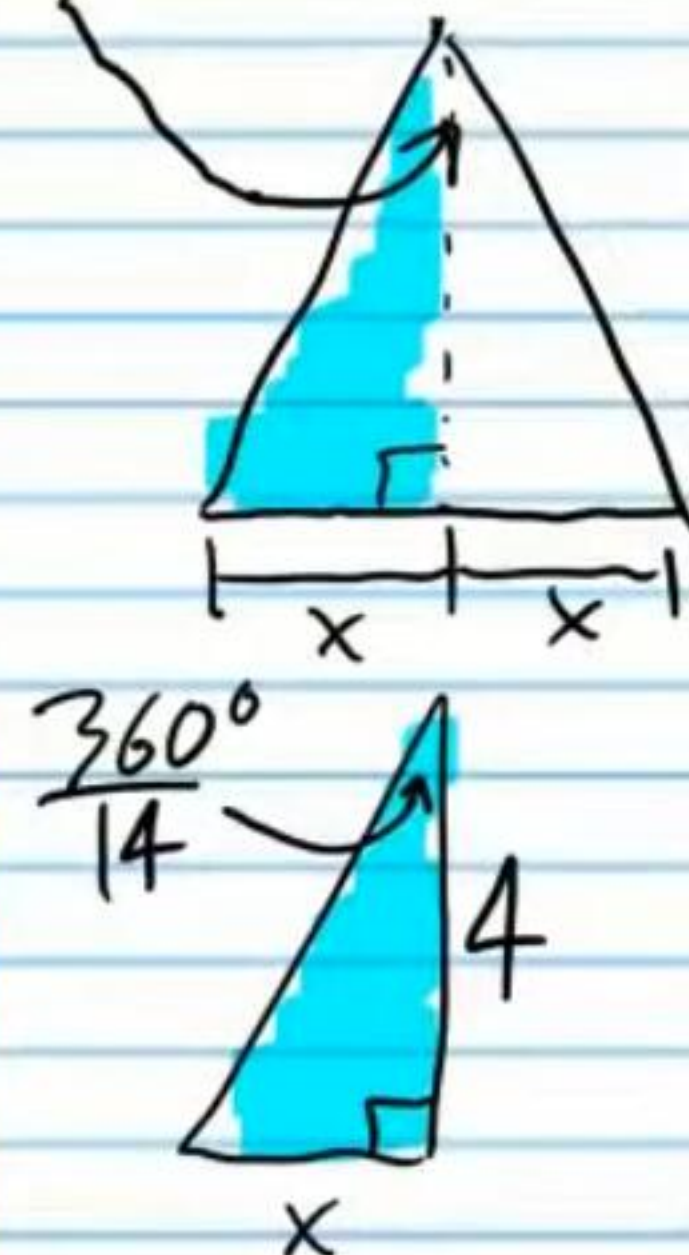
$$\tan\left(\frac{360^\circ}{14}\right) = \frac{x}{4}$$

$$x = 4\tan\left(\frac{180^\circ}{7}\right)$$

$$b = 2x$$

$$b = 2(4\tan\left(\frac{180^\circ}{7}\right))$$

$$b = 8\tan\left(\frac{180^\circ}{7}\right)$$



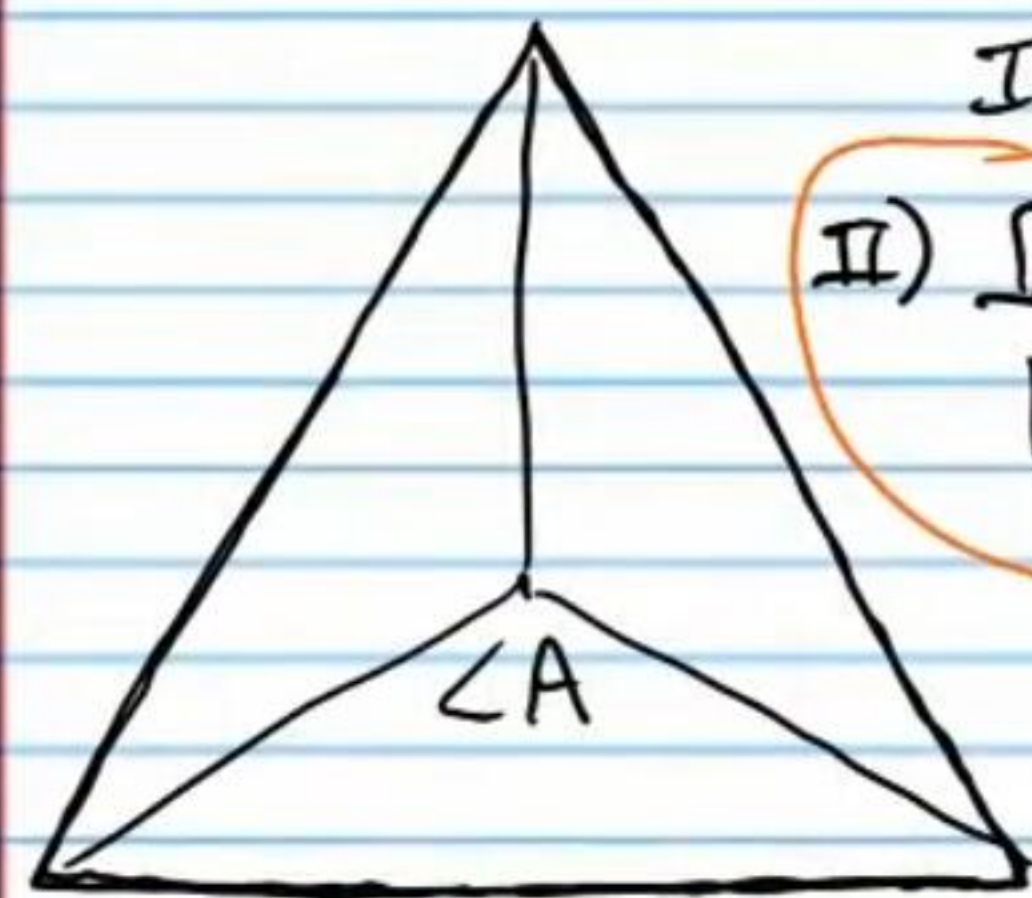
$$IV) A_{Tri} = \frac{1}{2}(8\tan\left(\frac{180^\circ}{7}\right))(4)$$

$$V) A_{Total} = 7A_{Tri}$$

$$A_{Total} = 7\left[\frac{1}{2}(8\tan\left(\frac{180^\circ}{7}\right))(4)\right]$$

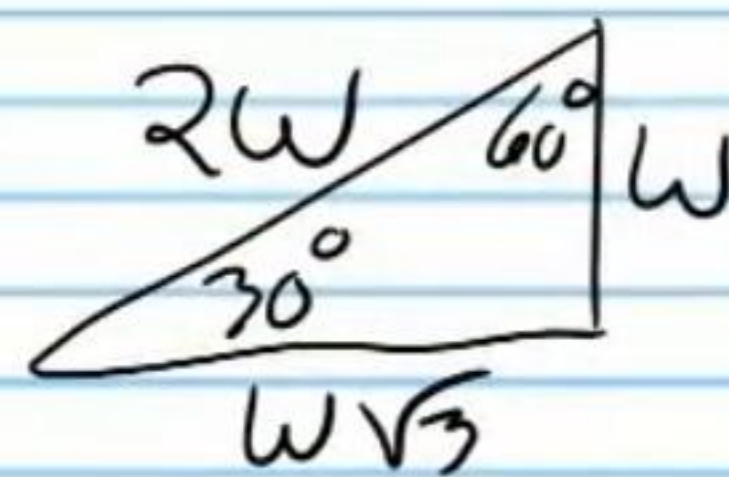
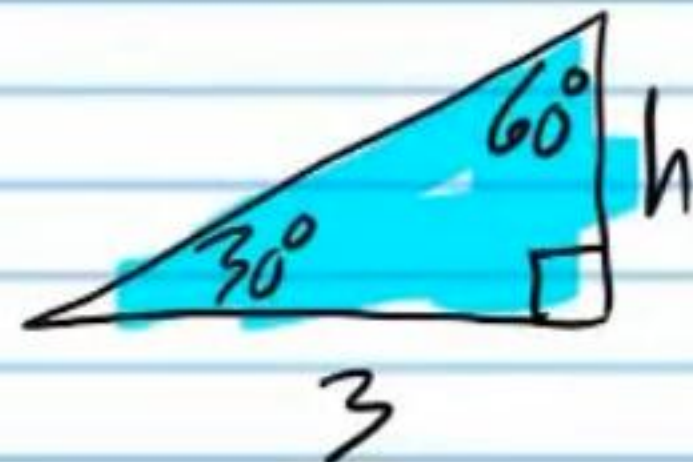
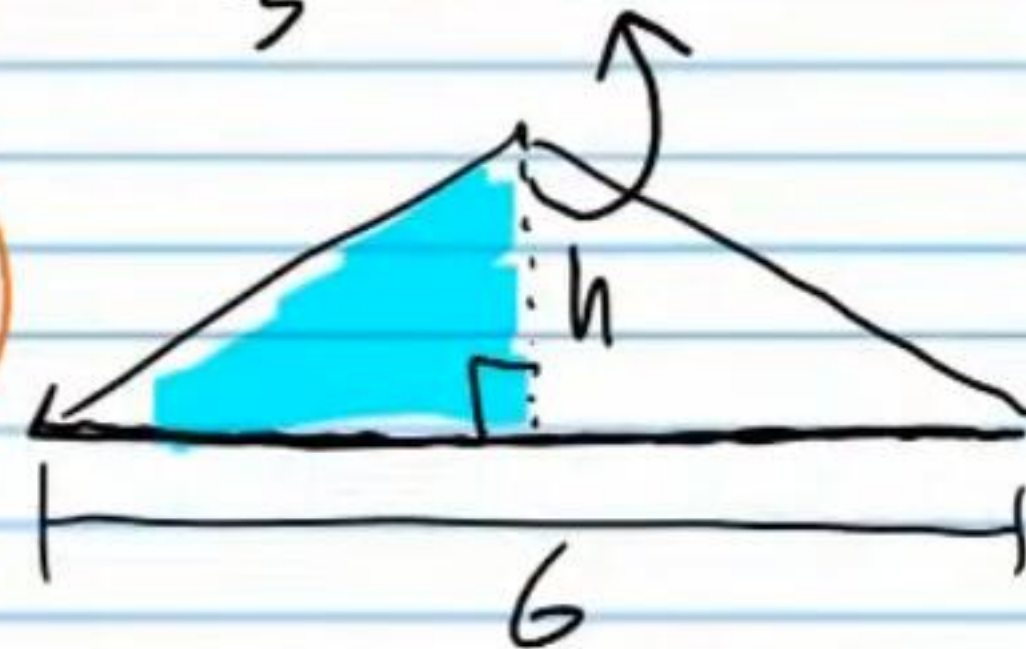
$$A_{Total} \approx 53.94 \text{ in}^2$$

⑥ Side Length: 6 cm



I) $m\angle A = \frac{360}{3} = 120^\circ$

II) Base
 $b = 6$



III) Height

$$W\sqrt{3} = 3$$

$$W = \frac{3}{\sqrt{3}}$$

$$h = W = \frac{3}{\sqrt{3}}$$

IV) $A_{Tri} = \frac{1}{2}(6)\left(\frac{3}{\sqrt{3}}\right)$

V) $A_{Total} = 3A_{Tri}$

$$A_{Total} = 3\left[\frac{1}{2}(6)\left(\frac{3}{\sqrt{3}}\right)\right]$$

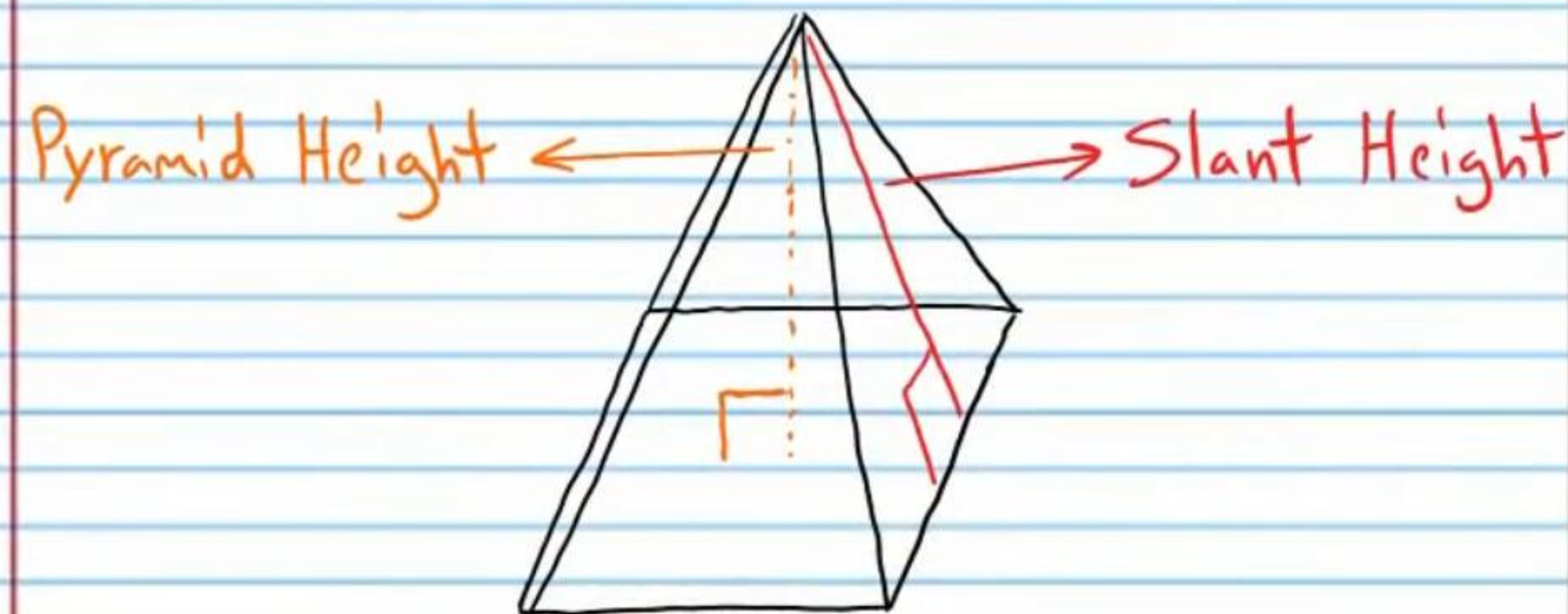
$$A_{Total} \approx 15.59 \text{ cm}^2$$

Surface Area of Regular Pyramids

Pyramid: A three dimensional shape with a polygon base and triangular sides with a common vertex at the top of the pyramid. The sides are called lateral faces.

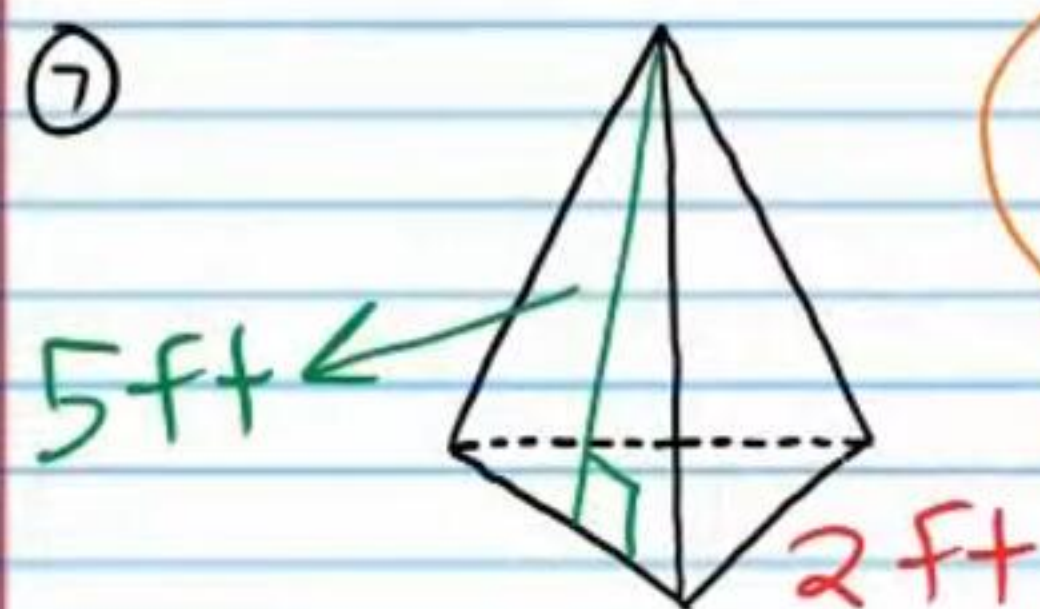
Regular Pyramid: A pyramid with a regular polygon for a base and with congruent isosceles triangles for sides. The segment connecting the center of the base and the vertex is perpendicular to the base and is called the height of the pyramid. The height of each side triangle is called the slant height.

Example: Square Base Regular Pyramid.



Find the surface area of each regular pyramid below.

⑦

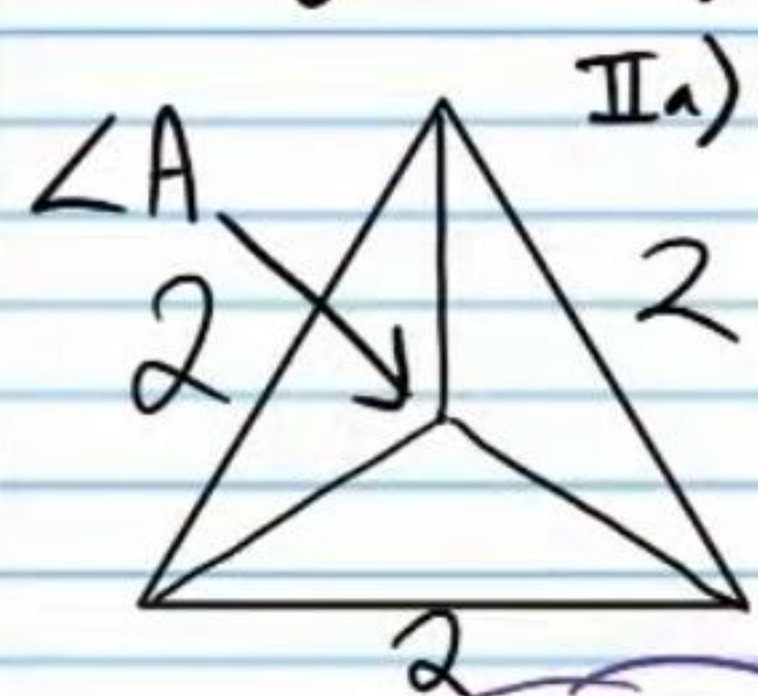


I) Lateral Faces

$$A_{\text{Tri}} = \frac{1}{2}(2)(5) = 5$$

$$A_{\text{sides}} = 3A_{\text{Tri}} = 3(5) = 15$$

II) Regular Polygon Base



$$\text{IIa) } m\angle A = \frac{360}{3} = 120^\circ$$

$$\text{II d) } A_{\text{Tri}} = \frac{1}{2}(2)\left(\frac{1}{\sqrt{3}}\right)$$

$$\text{IIe) } A_{\text{Polygon Base}} = 3A_{\text{Tri}}$$

$$A_{\text{Polygon Base}} = 3\left[\frac{1}{2}(2)\left(\frac{1}{\sqrt{3}}\right)\right] = \frac{3}{\sqrt{3}}$$

$$\text{III) } A_{\text{Pyramid}} = A_{\text{Polygon Base}} + A_{\text{sides}}$$

$$A_{\text{Pyramid}} = \frac{3}{\sqrt{3}} + 15 \text{ ft}^2$$

IIb) Height

$$1 = w\sqrt{3}$$

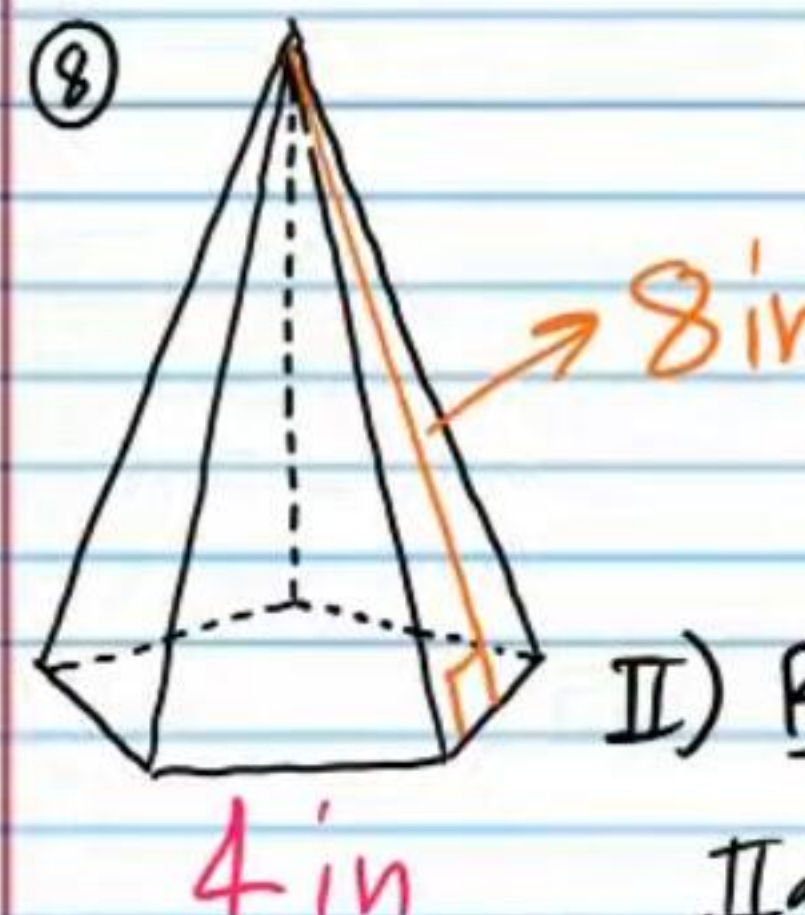
$$w = \frac{1}{\sqrt{3}}$$

$$h = w = \frac{1}{\sqrt{3}}$$

IIc) Base

$$b = 2$$

⑧



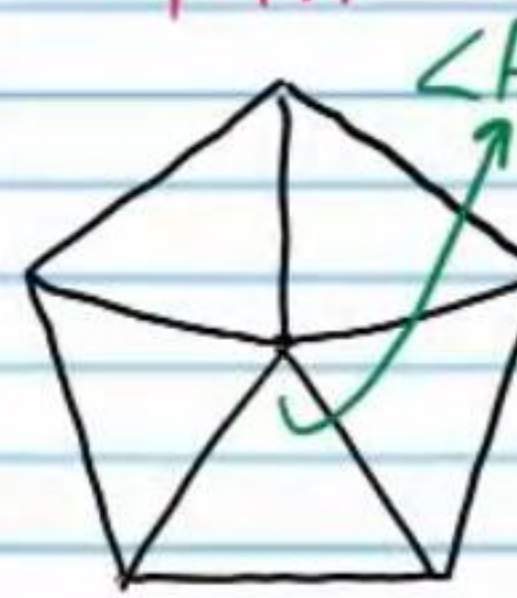
I) Lateral Faces

$$A_{\text{Tri}} = \frac{1}{2}(4)(8) = 16$$

$$A_{\text{sides}} = 5A_{\text{Tri}} = 5(16) = 80$$

II) Regular Polygon Base

$$\text{IIa) } m\angle A = \frac{360}{5} = 72^\circ$$



IIc) Base

$$b = 4$$

$$\text{II d) } A_{\text{Tri}} = \frac{1}{2}(4)\left(\frac{2}{\tan 36^\circ}\right)$$

$$\text{IIe) } A_{\text{Polygon Base}} = 5A_{\text{Tri}}$$

$$A_{\text{Polygon Base}} = 5\left[\frac{1}{2}(4)\left(\frac{2}{\tan 36^\circ}\right)\right]$$

$$A_{\text{Polygon Base}} \approx 27.5276$$

$$\text{III) } A_{\text{Pyramid}} = A_{\text{Polygon Base}} + A_{\text{sides}}$$

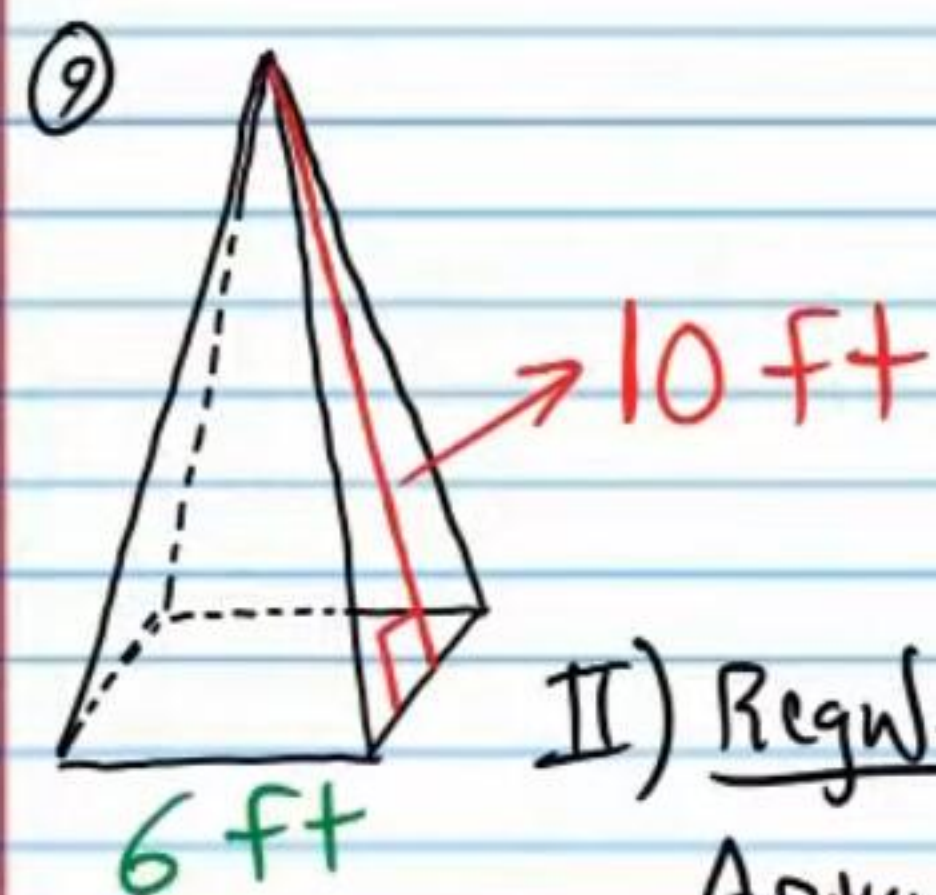
$$A_{\text{Pyramid}} \approx 27.5276 + 80$$

$$A_{\text{Pyramid}} \approx 107.53 \text{ in}^2$$

IIb) Height

$$\tan 36^\circ = \frac{2}{h}$$

$$h = \frac{2}{\tan 36^\circ}$$



I) Lateral Faces

$$A_{tri} = \frac{1}{2}(6)(10) = 30$$

$$A_{sides} = 4 A_{tri} = 4(30) = 120$$

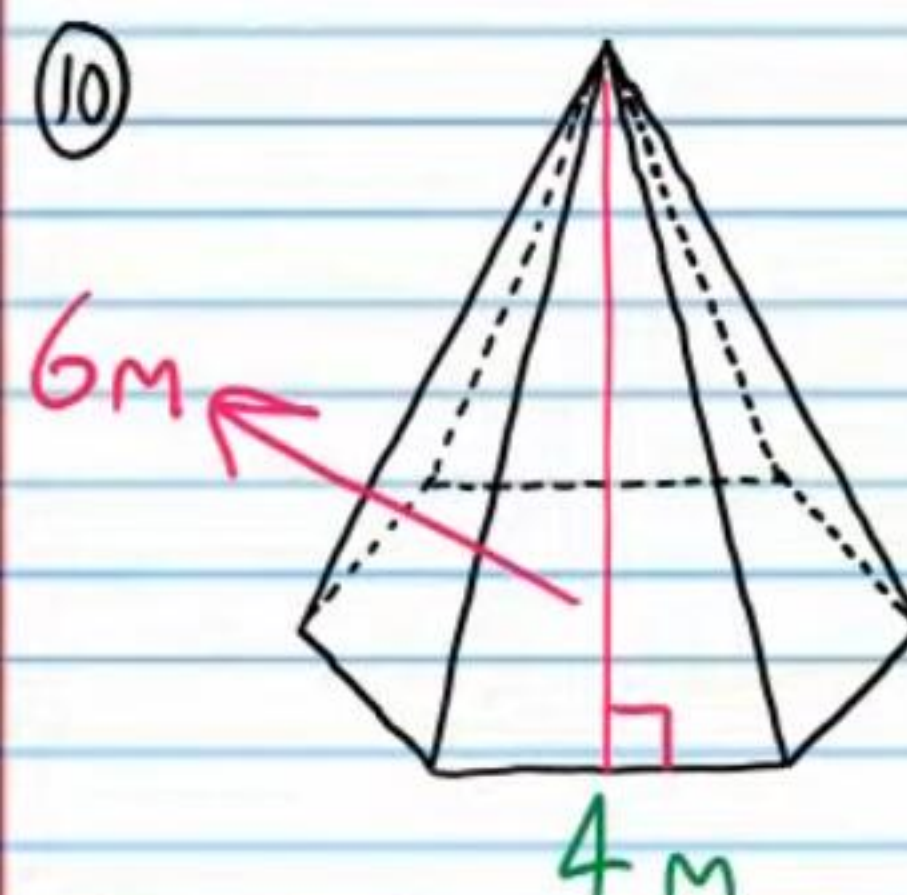
II) Regular Polygon Base

$$A_{Polygon Base} = 6(6) = 36$$

III) $A_{pyramid} = A_{Polygon Base} + A_{sides}$

$$A_{pyramid} = 36 + 120$$

$$A_{pyramid} = 156 \text{ ft}^2$$



I) Lateral Faces

$$A_{tri} = \frac{1}{2}(4)(6) = 12$$

$$A_{sides} = 6 A_{tri} = 6(12) = 72$$

II) Regular Polygon Base

IIa) $m\angle A = 360/6 = 60^\circ$



IIc) Base
 $b = 4$

II d) $A_{tri} = \frac{1}{2}(4)(2\sqrt{3})$

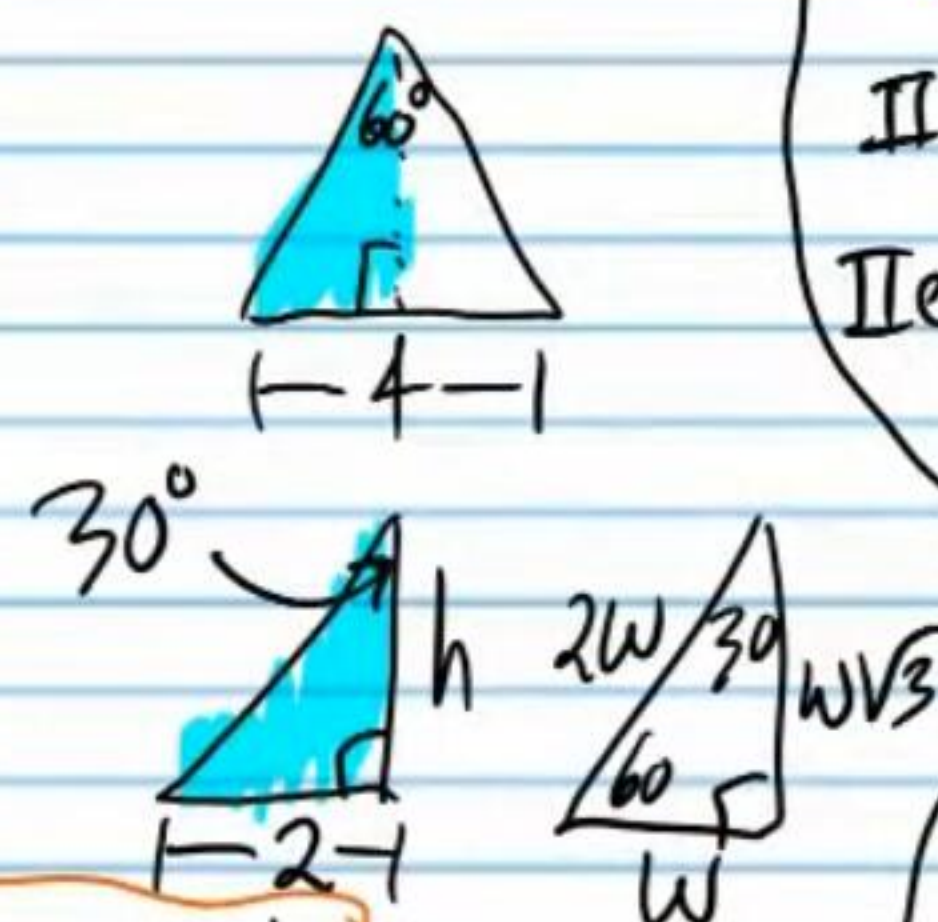
II e) $A_{Polygon Base} = 6 A_{tri}$

$$A_{Polygon Base} = 6 \left[\frac{1}{2}(4)(2\sqrt{3}) \right]$$

$$A_{Polygon Base} = 24\sqrt{3}$$

III) $A_{pyramid} = A_{Polygon Base} + A_{sides}$

$$A_{pyramid} = 24\sqrt{3} + 72 \text{ m}^2$$



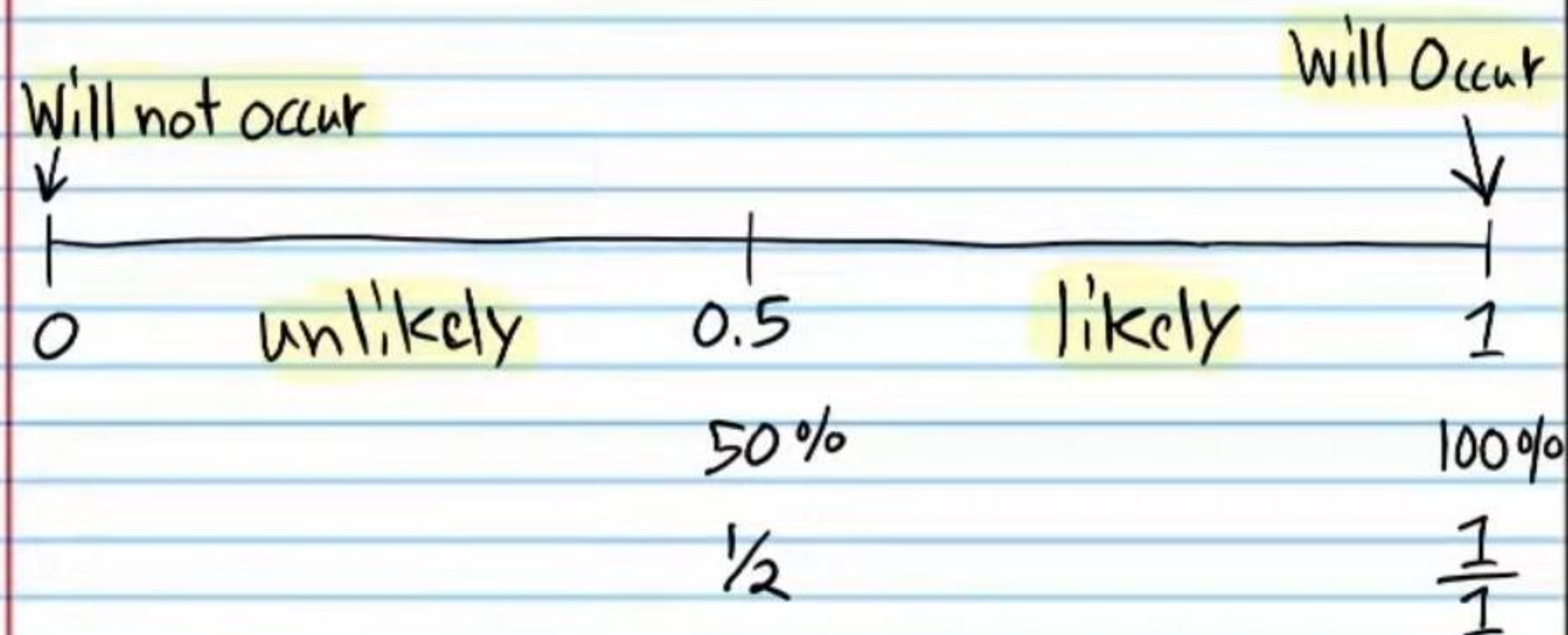
II b) Height

$$w = 2$$

$$h = w\sqrt{3} = 2\sqrt{3}$$

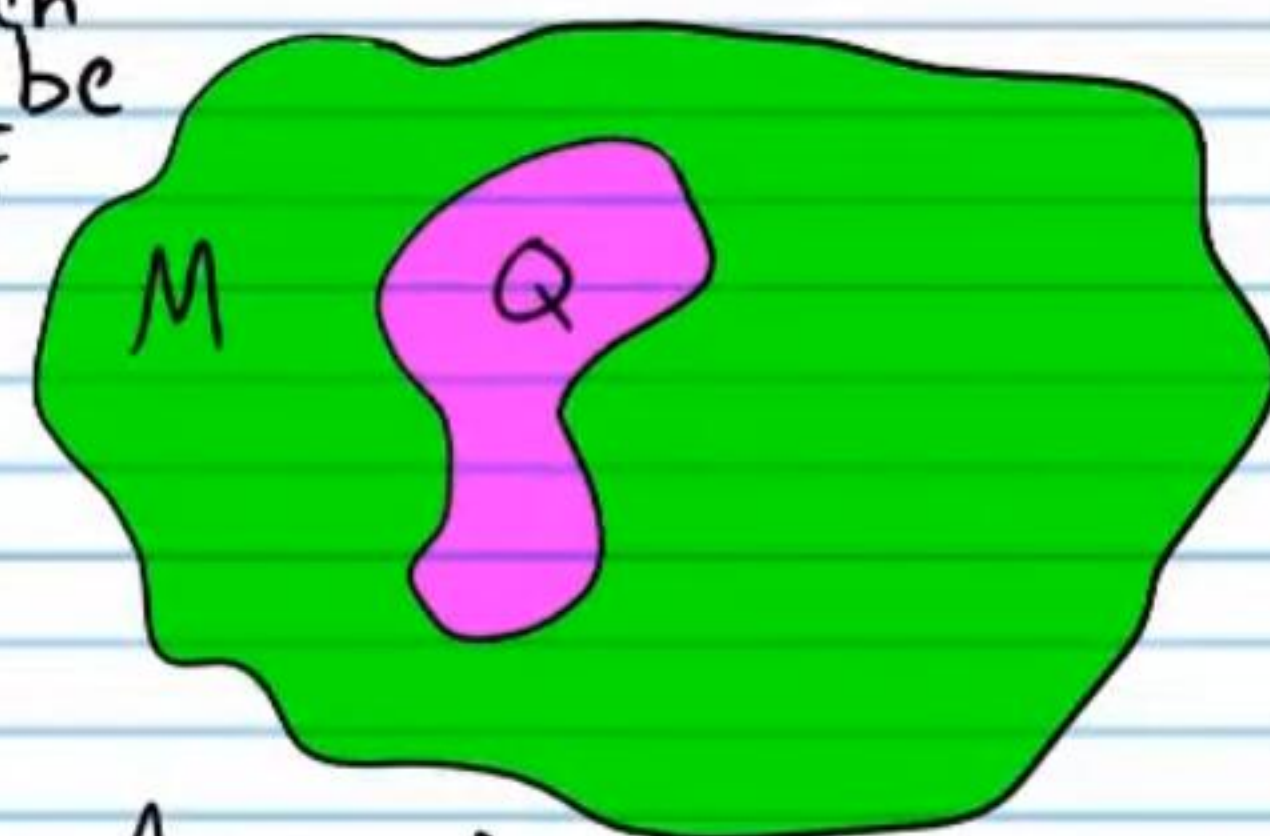
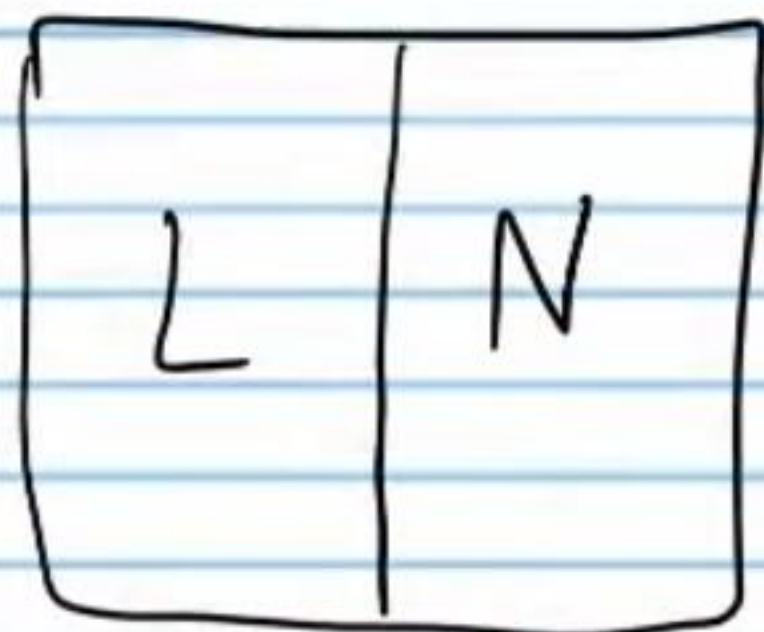
Geometric Probability

Probability: A number from zero to one that indicates the likelihood of an event.



Probability and Area

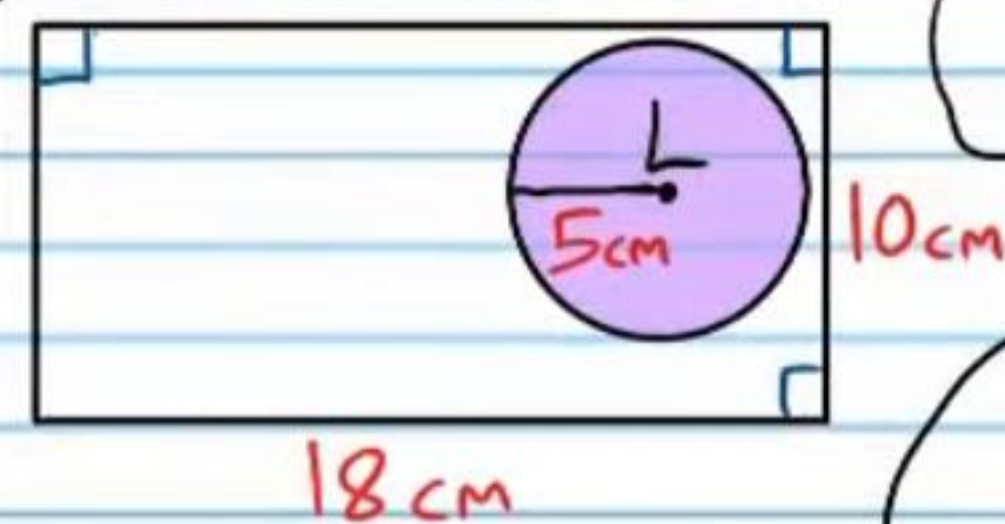
If region M contains region Q, then the probability that a randomly chosen point in M will also be in Q is the ratio of the area of Q to the area of M.



$$\frac{A_Q}{A_{L \& N}} = \frac{1}{2} = 50\%$$

Find the probability that a randomly chosen point in the figure will lie in the shaded region. Round to the nearest thousandth if necessary.

⑪ L is the center.

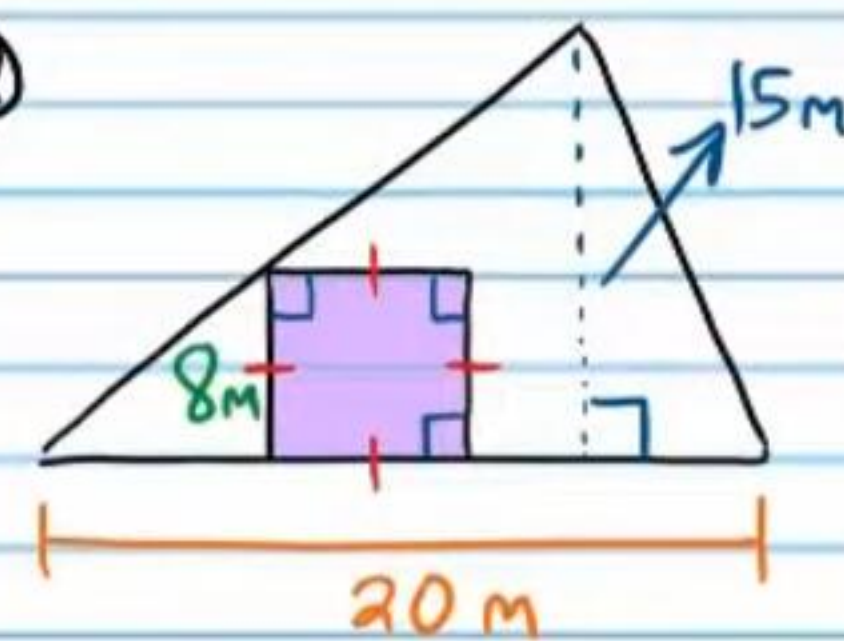


$$\begin{aligned} \text{I) } A_{\text{shaded}} &= A_{\text{circle}} \\ &= \pi(5)^2 \\ &= 25\pi \end{aligned}$$

$$\begin{aligned} \text{II) } A_{\text{Total}} &= A_{\text{rectangle}} \\ &= 18(10) \\ &= 180 \end{aligned}$$

$$\begin{aligned} \text{III) } P &= \frac{A_{\text{shaded}}}{A_{\text{Total}}} \\ P &= \frac{25\pi}{180} \\ P &\approx 0.436 \end{aligned}$$

⑫

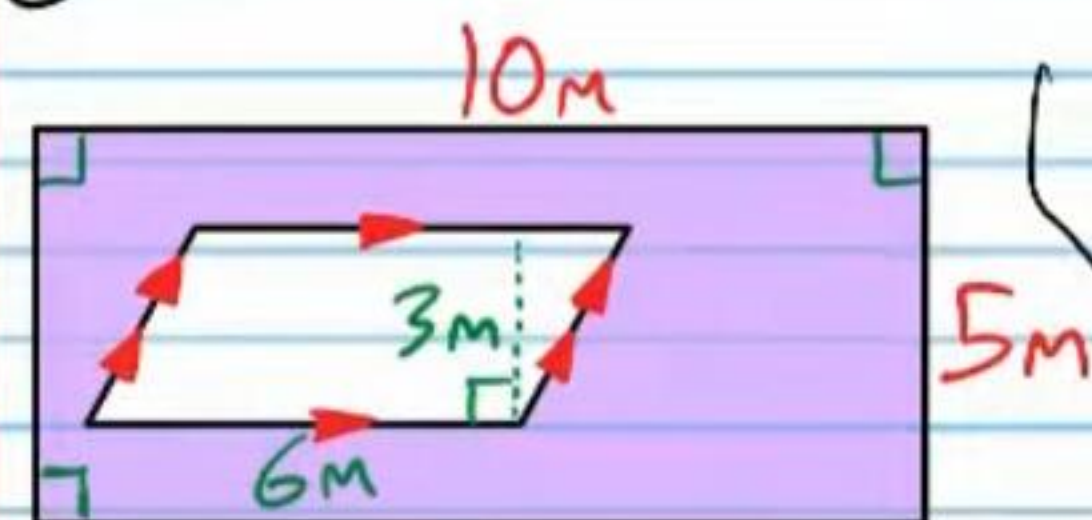


$$\text{I) } A_{\text{shaded}} = A_{\text{square}} = 8(8) = 64$$

$$\begin{aligned} \text{II) } A_{\text{Total}} &= A_{\text{tri}} \\ &= \frac{1}{2}(20)(15) \\ &= 150 \end{aligned}$$

$$\begin{aligned} \text{III) } P &= \frac{A_{\text{shaded}}}{A_{\text{Total}}} \\ &= \frac{64}{150} \\ &\approx 0.427 \end{aligned}$$

13



$$I) A_{\text{shaded}} = A_{\text{rectangle}} - A_{\text{para}}$$

$$A_{\text{shaded}} = 10(5) - 6(3)$$

$$A_{\text{shaded}} = 50 - 18$$

$$= 32$$

$$II) A_{\text{Total}} = A_{\text{rectangle}}$$

$$= 10(5)$$

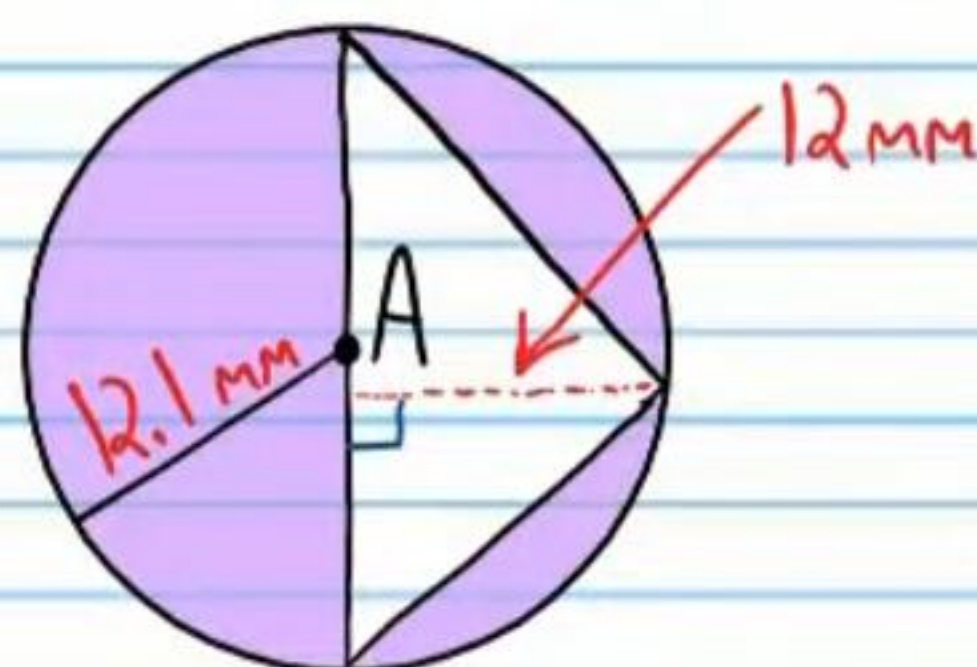
$$= 50$$

$$III) P = \frac{A_{\text{shaded}}}{A_{\text{Total}}}$$

$$= \frac{32}{50}$$

$$= \boxed{0.64}$$

14 A is the center.



$$I) A_{\text{shaded}} = A_{\text{circle}} - A_{\text{tri}}$$

$$A_{\text{shaded}} = \pi(12.1)^2 - \frac{1}{2}(12.1 + 12.1)(12)$$

$$\approx 459.96058 - 145.2$$

$$\approx 314.76058$$

$$II) A_{\text{Total}} = A_{\text{circle}} = \pi(12.1)^2$$

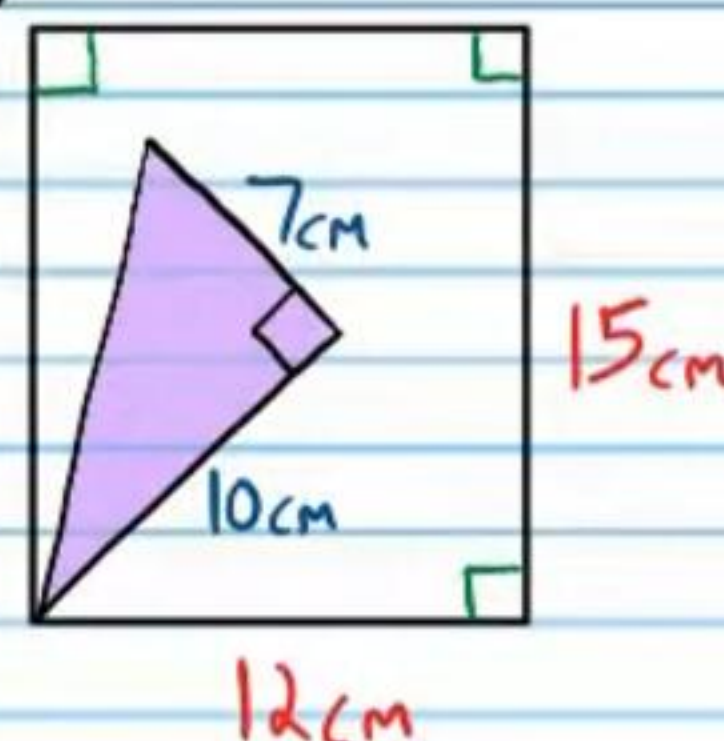
$$\approx 459.96058$$

$$III) P = \frac{A_{\text{shaded}}}{A_{\text{Total}}}$$

$$= \frac{314.76058}{459.96058}$$

$$\approx \boxed{0.684}$$

15



$$I) A_{\text{shaded}} = A_{\text{tri}}$$

$$= \frac{1}{2}(10)(7)$$

$$= 35$$

$$II) A_{\text{Total}} = A_{\text{rectangle}}$$

$$= 15(12)$$

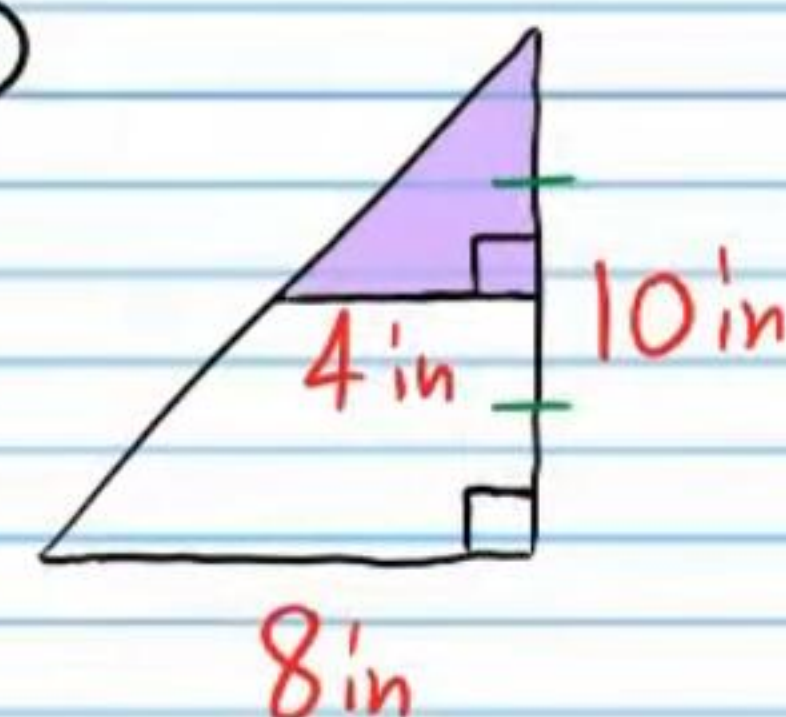
$$= 180$$

$$III) P = \frac{A_{\text{shaded}}}{A_{\text{Total}}}$$

$$P = \frac{35}{180}$$

$$\boxed{P \approx 0.194}$$

16



$$I) A_{\text{shaded}} = A_{\text{top Triangle}}$$

$$= \frac{1}{2}(4)(5)$$

$$= 10$$

$$II) A_{\text{Total}} = A_{\text{Big Triangle}}$$

$$= \frac{1}{2}(8)(10)$$

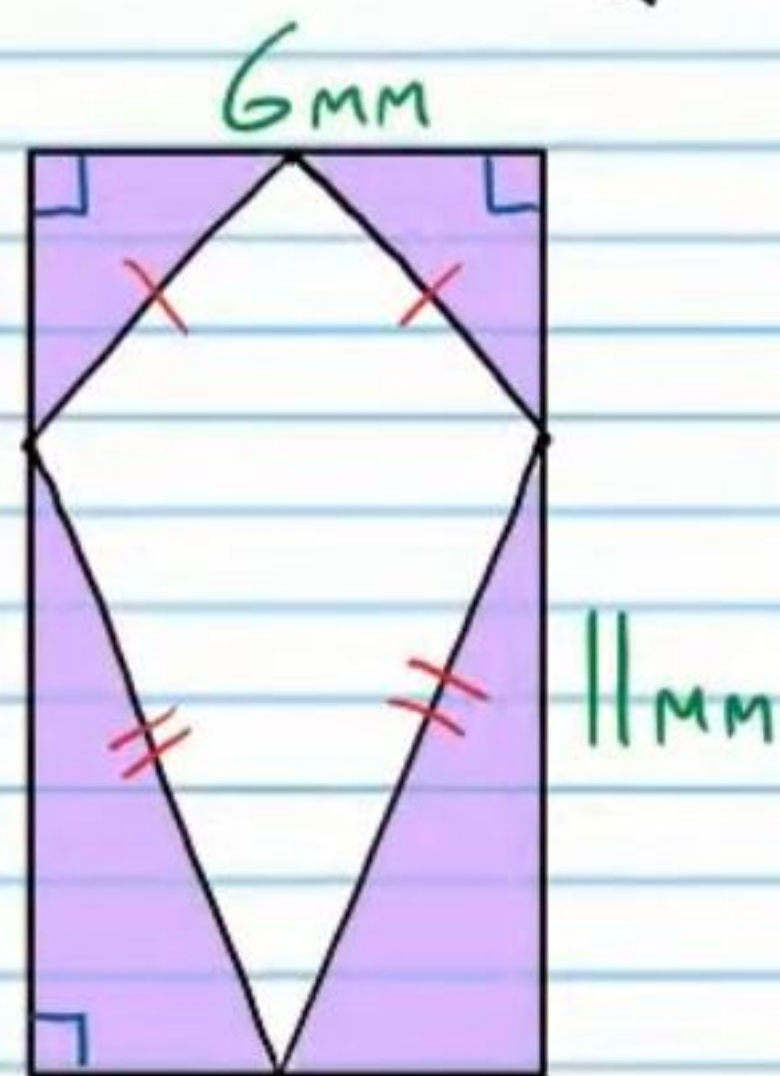
$$= 40$$

$$III) P = \frac{A_{\text{shaded}}}{A_{\text{Total}}}$$

$$= \frac{10}{40}$$

$$= \boxed{0.25}$$

①7 Remember, the area of a kite is $\frac{1}{2}d_1d_2$.

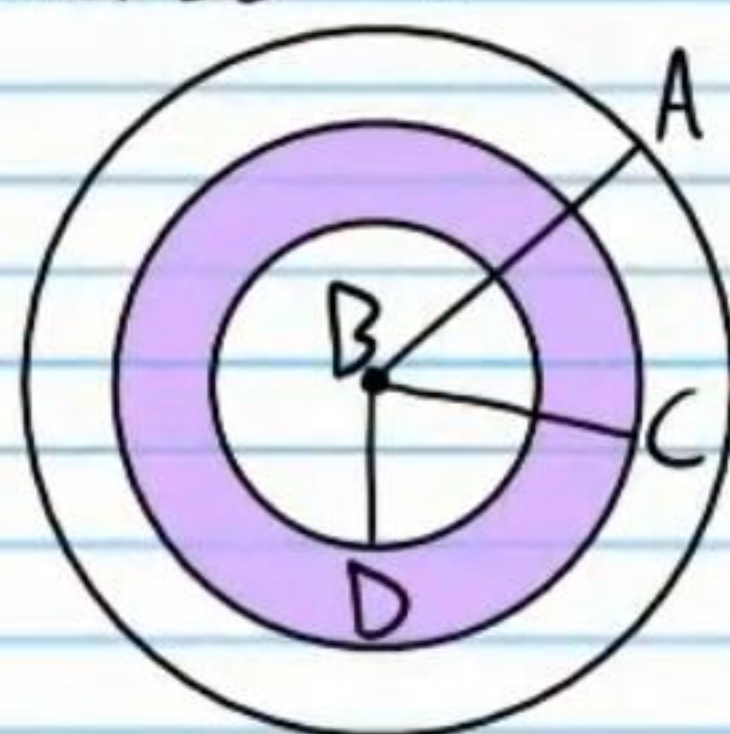


$$\begin{aligned} \text{I) } A_{\text{shaded}} &= A_{\text{rectangle}} - A_{\text{kite}} \\ &= 6(11) - \frac{1}{2}(6)(11) \\ &= 66 - 33 \\ &= 33 \end{aligned}$$

$$\begin{aligned} \text{II) } A_{\text{total}} &= A_{\text{rectangle}} \\ &= 6(11) \\ &= 66 \end{aligned}$$

$$\begin{aligned} \text{III) } p &= \frac{A_{\text{shaded}}}{A_{\text{total}}} \\ &= \frac{33}{66} = \boxed{0.5} \end{aligned}$$

①8 $AB = 10$, $BC = 7$
and $BD = 4$.



$$\begin{aligned} \text{I) } A_{\text{shaded}} &= A_{\text{circle BC}} - A_{\text{circle BD}} \\ &= \pi(7)^2 - \pi(4)^2 \\ &= 49\pi - 16\pi \\ &= 33\pi \end{aligned}$$

$$\begin{aligned} \text{II) } A_{\text{total}} &= A_{\text{circle AB}} \\ &= \pi(10)^2 \\ &= 100\pi \end{aligned}$$

$$\begin{aligned} \text{III) } p &= \frac{A_{\text{shaded}}}{A_{\text{total}}} \\ &= \frac{33\pi}{100\pi} \\ &= \frac{33}{100} \\ &= \boxed{0.33} \end{aligned}$$